

“LET THE WILD IN”:

FERMENTATION, CLEANLINESS
AND COMPETITIVE COLLABORATION IN MULTISPECIES
BREWERY

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THESIS

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Abstract

This thesis explores the dynamic relationships between humans and microbes through an ethnographic study of a natural brewery in Yunnan, China. Focusing on fermentation as both a material practice and a relational process, it asks: what kind of relationship forms between brewers and yeast, and how does fermentation itself create the space for that relationship? Drawing on multispecies ethnography and feminist epistemology, I develop the concept of competitive collaboration to describe how brewers and microbes engage in a continuous, non-verbal negotiation, especially over the meaning and maintenance of cleanliness. Through participant observation and embodied practice, I show that fermentation is not a process of full human control, nor of microbial freedom. Instead, it is a shared space shaped by friction, care, and adjustment. Cleanliness here is not a fixed standard but a contested category, shaped by human intention and microbial behavior. I argue that knowledge in this space is situated, sensory, and relational: it emerges not through mastery, but through ongoing attentiveness to both visible and invisible signs. By foregrounding this competitive collaboration, the thesis contributes to current debates in anthropology about nonhuman agency, human and nonhuman relationship, and the politics of interspecies knowledge-making. Fermentation becomes a method for rethinking how knowledge, responsibility, and coexistence are practiced, not only among humans, but across species boundaries.

Introduction: Rethinking Knowledge through Multispecies Entanglements

In recent decades, anthropology has experienced a shift toward reimagining human-nonhuman relations. Moving beyond the view of nature as a passive background, multispecies ethnography asks what it means to live, work, and think with other forms of life (Haraway, 2008; Tsing, 2015). It invites anthropologists to attend not only to the human world, but to the entangled, co-constituted lifeworld of animals, plants, fungi, and microbes (van Dooren, 2011; Paxson, 2013; Tsing, 2015). This perspective challenges the traditional divide between culture and nature, subject and object, and creates space for new forms of ethnographic attention. Within this shift, fermentation emerges as a particularly rich field for anthropological inquiry. Just like Maya Hey says, it is a practice where human and microbial lives literally meet and transform one another (Hey, 2021). It is also a domain where agency is not exclusive to humans. Instead, it is distributed, negotiated, and sometimes resisted (Metcalf, 2011). I believe, fermentation invites us to ask: who acts? who knows? who decides what counts as clean, healthy, or alive?

This thesis builds on these questions through an ethnographic study of a small natural brewery in Mile, Yunnan, China. Mile is located in the mountainous southeast of China, a region where multiple ethnic groups live and work. The area spans elevations from 300 to 3,000 meters, with a subtropical climate that supports a wide range of species and habitats. I joined the brewery team as an assistant brewer and participated in every step of the craft beer production process. I also contributed to designing the brewery's brand logo and labels and supported their campaign on social media. The brewers came from a mix of local ethnic groups and other cities. Out of respect for their wishes, I do not include any personal information or names in this study. They all had received formal training in brewing schools and viewed brewing as both a profession and

a livelihood. Some of them had studied traditional techniques in Germany and later localized these practices by integrating natural features and sensory knowledge from Yunnan, especially in the fermentation process. As one brewer told me, “Fermentation is the life of beer.” During my fieldwork, I participated in the brewing process, with a particular focus on fermentation. I became attentive not only to the technical aspects of brewing, but also to the ways in which brewers and yeast interacted, guided by timing, temperature, containment, and intuition. From these observations emerged my central research question: what kind of relationship forms between brewers and yeast through fermentation? And, more specifically, how does fermentation itself create the space for that relationship to occur?

In this research, I describe the relationship between brewers and microbial agents as competitive collaboration. It is a dynamic, non-verbal engagement in which both brewers and yeast silently, yet persistently, struggle over the definition and control of cleanliness. Fermentation mediates this struggle. It becomes not only a chemical process, but also a space of negotiation, a kind of silent dialogue that crosses species boundaries. Within this space, decisions about what counts as “clean,” “wild,” or “safe” are not made once and for all, but are continuously produced and adjusted.

To make sense of this relationship and this space, I draw on two major bodies of theory: multispecies ethnography and feminist epistemology. These fields have been developed in different contexts, but both share a commitment to rethinking knowledge: how it is produced, who participates, and what forms of life are recognized as knowing (Haraway, 1988, 2016; Kafer, 2013). Fermentation becomes a point of convergence between these perspectives. It is both a material practice and a metaphor, a daily activity as well as a theoretical lens. In what

follows, I trace how anthropologists and feminist scholars have explored human-nonhuman relations through concepts such as entanglement, companion species, symbiosis, and situated knowledge (Zwart, 2000; Haraway, 2008, 1988). I then show how fermentation extends and complicates these concepts, especially when scaled down to the microbial level. While much multispecies research has focused on plants, animals, and insects, beings whose presence can be seen and named, microbial life requires different modes of attention. It challenges the ethnographer to listen differently, to wait, and to interpret signs that are often ambiguous, shifting, and embodied.

It is precisely this kind of knowledge that fermentation brings to the fore. As I will show, brewing involves collaboration, but not in a romantic or harmonious way. It is to enter a competitive, uncertain, and deeply material relationship with beings that do not speak, but act. The yeast does not obey. It transforms. It resists. It surprises. And it does so within a space shaped by human history, scientific regulation, cultural norms, and embodied labor. This thesis sets the stage for a theoretical framework that treats fermentation as more than technique. It is a site of epistemological tension, where questions of control, cleanliness, and collaboration come to the surface, sometimes quite literally. By attending to this tension, and by placing it in dialogue with feminist and multispecies theory, I aim to contribute to current anthropological debates about how knowledge is made, not only by humans but also in relation with other forms of life.

Methodology: Embodied Practice and Situated Positionality

In the process of brewing, the brewer must learn to adjust themselves according to the environment and everything unfolding around them; success depends on adapting to the

experiential dynamics of the process. During the three months I spent participating in brewing and related practices, I immersed myself fully in becoming a brewer. Beyond participant observation and interviews with brewers, I learned through practice, making myself an inseparable part of the brewing process. This inseparability was primarily reflected in my effort to read into the experiences of each nonhuman participant, those who could not express themselves through human language, and to connect these with my own experiences. As Maya Hey (2021) suggests, through specific sensory engagements and immersive participation, one can decenter human subjectivity by practicing an attention to nonhuman agencies. I applied this kind of embodied practice directly during my fieldwork—it allowed me to better understand what the silent beer foam, subtle shifts in color, and unusual smells were trying to express.

Autoethnography enabled me to naturally integrate the inescapable subjectivity of qualitative methods, offering a lens through which I could better perceive the dynamic changes in relationships between human and nonhuman participants during the research process (Pink, 2009). Autoethnography is understood as a method that draws on the researcher's personal experience to interpret and understand the cultural expressions, beliefs, and actions encountered in the field (Pink, 2009; Pintér, 2024). In addition to incorporating subjectivity as part of the narrative, I believe this perspective also allows for the visibility of the researcher's identity and positionality and the experiential associations that may connect them to the research content.

As a Mandarin-speaking Han Chinese researcher, I encountered linguistic barriers in this multi-ethnic, mountainous region of southwestern China. The only language I could use was Standard Mandarin, and sadly, I did not speak any other dialect or language. Here, I want to emphasize the term “language” rather than “dialect,” because in Mile, most local people are Hani

and Yi. The Hani speak Hani language, which belongs to the Yi branch of the Tibeto-Burman language family. The Yi speak Yi language, from a different branch of the same family. For me, both were completely difficult—I had no clue what was being said.

Since the mid-1950s, the Chinese government has promoted the use of Mandarin nationwide. As a result, most people, including elders, can understand Mandarin, though some still find it difficult to speak fluently. Fortunately, everyone I worked with under the age of fifty spoke fluent Mandarin, which made communication significantly easier. Still, my difference remained visible. When people around me began speaking in their native languages, I felt like a creature from outer space—human-like in appearance, yet entirely incapable of understanding. When I spoke, my words were always public. My language had no privacy; everyone knew exactly what I was saying.

This sense of unfamiliarity made me cautious at the beginning of my work in the brewery, as I feared that unfamiliarity might gradually turn into a gaze, into a form of objectification that could solidify into irreversible stereotypes. This concern stems from my position as part of the ethnic majority in a large nation. I am Han Chinese, a group that, according to China's 2021 national census, constitutes approximately 91% of the population, while the remaining 55 officially recognized minority groups collectively make up only 9% (National Bureau of Statistics & Office of the Leading Group of the State Council for the Seventh National Population Census, 2021). But Yunnan has the highest population of ethnic minorities in China. The interwoven histories, languages, and cultures shaped over centuries here make this region markedly different from others. Still, this sense of belonging to the same country allowed me to avoid exoticizing the local context. It gave me a way to step back and

reflect on cultural practices without positioning myself entirely as an outsider. We still share many overlapping experiences.

This inside-outsider perspective allowed me to integrate my own narrative into the broader narrative of my research. These seemingly contradictory positionalities existed and fused in unprecedented ways. The term inside outsider was introduced by Patricia Hill Collins (1990, 2022) in her discussion of black female intellectuals navigating the margins of academia. It refers to a knowledge making standpoint forged through multiple identity positions. When I tried to apply this feminist methodological insight to my own research, I found it helpful in two distinct ways.

First, it challenged me to reflect on how to resist the power-laden gaze that may arise when a researcher comes from a dominant social position, and more broadly, how to listen sincerely to the voices of nonhuman participants when humans are the primary producers of discourse. The former connects directly to my own positionality: I cannot detach from my identity, since I was never born into a condition of otherness. The best way to resist this perspective is to relate personal experience to knowledge production in the field. For instance, while observing the brewing process of a local drink called sweet white wine(甜白酒), I noticed that the technique resembled a lightly fermented food my mother used to make at home. Using this connection as a starting point, I initiated conversations with the drink-craft lady and asked questions about their process. This shared experiential reference facilitated our interaction and reduced initial tension, allowing dialogue to begin in a more reciprocal space. When I extended this approach to encounters with nonhuman actors, it became more like a form of reading, one that involved attending to what was being revealed and observing accordingly. This resembles

what Anna Tsing (2015) calls the art of noticing: a form of highly concentrated attention directed toward tiny life forms. If we must name this with an equally abstract and intense term, it might be a gesture toward de-centering the human.

Second, the contribution of feminist methodology lies in its focus on seeing individuals within relationships, rather than treating them as passive data sources for extraction (Skeggs, 2001 ; Hey, 2021). As I navigated my own complex positionality, I came to realize that this personal dimension also mirrored the relational patterns within the small-scale brewery, especially in human-multispecies interactions. Humans rely on socially constructed identity cues such as names, clothing, language, and accents to assess one another. These ambiguous markers form the boundaries of classification systems. But can such layered categorization be applied to nonhuman others, such as yeasts, microbes, and agents without stable definition? Perhaps not easily. I believe that my own inability to clearly distinguish my social position from those of other human workers in the brewery made it possible for me to see how yeast and other nonhumans maintained legible, repeatable relationships in the midst of indeterminacy. Fieldwork, in this case, became a mirror. The multispecies environment reflected my own conditions back to me. I could see the blurred boundaries between species because my relationships with other humans were just as fluid.

As Jack Katz (2018) puts it: your problem is their problem. I began to feel that my own relationship with the local communities in this land mirrored, in some way, the relationship between microbes and brewers in this space. It may not have been a “problem” in the negative sense, but it was certainly a powerful way to think about how relationships entangle the researcher with those they study. Relationships are dynamic, but the captured moment of

interaction can be a highly condensed frame with the researcher's own voice embedded.

Throughout this process, I held multiple roles including assistant brewer, creative lead, the only woman in the brewery, anthropologist, and a Mandarin-speaking northerner. I found that my attempts to build relationships revealed how multispecies actors also established connections.

These actors took many forms: silent powders, fermentation starters, microbial communities that shaped flavor, invisible agents determining the final brew. No one, human or nonhuman, could be entirely sure of who they were in the process. That uncertainty, it turns out, is shared.

Theoretical Framework

Microbial Life and the Post-Pasteurian Turn

While much of multispecies ethnography has concentrated on visible life forms such as plants, animals, and insects, the study of microbial life shifts anthropological attention to a different scale (Kirksey & Helmerich, 2010; Smart, 2014). Microbes are invisible to the naked eye, often nameless, and historically cast as threats rather than collaborators (Haraway, 2008; Ironstone, 2019; DeSalle & Perkins, 2015). Yet their presence is unavoidable. They live in our bodies, on our surfaces, and in our food (van Dooren, 2011; Tsing, 2015; Hey, 2021). In fermentation, they are the agents of transformation, making bread rise, beer bubble, and soy ferment (Hey, 2021, Pintér, 2024). To engage with microbes ethnographically is to confront the limits of human perception and the assumptions embedded in dominant knowledge systems.

One of the most influential responses to microbial life in anthropology comes from Heather Paxson's (2013) work on artisanal cheese production in the United States. Paxson introduces the concept of the post-Pasteurian, referring to a way of thinking and practicing that

moves beyond the strict hygienic logic of Louis Pasteur's legacy. The "Pasteurian" regime, as she explains, treats microbes as dangerous contaminants. It justifies sterilization, regulation, and control through scientific means (Metcalf, 2011). In contrast, post-Pasteurian producers do not try to eliminate microbial life—they learn to live with it.

In the small Yunnan brewery where I conducted fieldwork, I observed a similar logic. The brewers, though not using the language of "post-Pasteurianism," had a comparable orientation toward yeast. They worked with live, wild cultures. These microbes were not seen as mere risks, but as collaborators—necessary, unpredictable, and lively. Fermentation was not a process to control perfectly, but one to navigate, to negotiate, and often to respond to.

This negotiation echoes Foucault's (1978) concept of biopower, the ways in which life is governed not only through direct discipline but also through subtle forms of regulation, expertise, and moral framing. In Paxson's work, and in my own fieldsite, microbial life is not simply "free" under post-Pasteurian logic. It is still subject to forms of governance through temperature charts, fermentation times, health inspections, and aesthetic standards. What changes is the ethics of engagement. The aim is no longer to dominate microbes but to cohabitate with them under terms that are constantly shifting. This shift gives rise to what Paxson calls "microbial politics," a mode of thinking about how microbial life is valued, controlled, and made meaningful. These politics are practiced daily in fermentation spaces through waiting, tasting, discarding, and adjusting. They raise questions about risk, expertise, and the body. In these interactions, we can echo what Hub Zwart (2000) describes as a form of symbiosis, not only in the biological sense but also in material, sensory, and symbolic dimensions. Fermentation is not just about microbes doing chemical work. It is a shared practice. It requires humans to touch,

smell, and feel the process, and to make judgments without clear rules. As Hub Zwart (2000) suggests, this symbiosis is unstable. It can go wrong. It requires vigilance, patience, and care. These are not simply technical tasks. They are relational acts.

This relational aspect is beautifully illustrated in Maya Hey's (2019) ethnographic study of sake making. She argues that the brewers' knowledge is not only technical, but also moral and attunement (Hey, 2021). Sake brewer does not control fermentation in a top-down way. Rather, they wait for signs from the microbial world. They smell the barrels, observe the bubbles, and learn to "listen" to yeast. This kind of knowledge is bodily, emergent, and context specific. It depends on being present, on responding rather than predicting. In Hey's words, fermentation becomes a form of attunement: one that moves beyond human intention to include nonhuman agency (Hey, 2021).

These insights resonate deeply with my own fieldwork. In the brewery, I watched brewers spend hours adjusting small variables, opening a window for more airflow, cooling a barrel with wet towels, or simply waiting another day. These were not standardized practices. They were acts of engagement with microbial signals. But such engagement did not always look harmonious. The yeast could "rebel": fermenting too fast, producing off-flavors, or refusing to activate. The brewers, in turn, adapted—but they also pushed back. They discarded some batches. They sanitized equipment. They tried to guide the yeast, but they could never fully control it.

It is in this tension that my concept of competitive collaboration begins to take shape. The brewer and the yeast are not in conflict or harmony, but in a situation of silent negotiation. Each act in ways that influence the other. Each attempt to define the terms of "cleanliness," and

“quality.” This mutual influence, while maintaining distinct boundaries in the definition of cleanliness, echoes what Mandy de Wilde, Weera Koopman, and Annemarie Mol (2020) describe in their article: “one clean is not the other.” They discuss how the widespread adoption of face masks during the COVID-19 pandemic, intended to protect humans from viral intrusion, ironically introduced massive amounts of urban waste and oceanic plastic pollution. Here, cleanliness aimed at shielding the human body from microbes stands in contradiction to cleanliness understood as the protection of ecosystems from toxic contamination. Thus, it becomes crucial to consider multiple, co-existing forms of cleanliness and the tensions between them.

The cleanliness practiced inside the brewery is a necessary step for the brewer to ensure that the beer becomes an acceptable product. However, for the yeast, this same act of cleaning may pose a threat of death, as the sanitized environment created by cleaning agents is inhospitable to their survival. Their interaction is continuous, responsive, and full of friction. Crucially, this relationship unfolds without language. It is mediated by temperature, foam, smell, and time. It is a non-verbal conversation shaped by the limits of control and the presence of difference. Cleanliness becomes a contested space. No longer a fixed scientific ideal, but a site of struggle and communication. Fermentation, then, is not only a process of transformation. It is a material space where power, knowledge, and agency are co-produced between species.

Fermentation and Feminist Ways of Knowing

If multispecies ethnography invites us to attend to life beyond the human, shifting our attention to the entangled worlds of microbes, animals, and plants (Kirksey & Helmreich, 2010; van Dooren, 2011), feminist theory challenges us to question the foundations of knowledge

itself: who is allowed to know, how knowledge is produced, and whose perspectives are valued (Haraway, 1988; Puig de la Bellacasa, 2012). In both fields, there is a shared commitment to challenging dominant epistemologies—those based on distance, objectivity, and universalism. Instead, they propose situated, relational, and embodied forms of knowing-knowledge that emerges through contact, care, and context. At the heart of feminist epistemology is Donna Haraway's (1988) concept of situated knowledge. Haraway argues that all knowledge is partial, located, and shaped by the position of the knower. Every observation comes with its own conditions and responsibilities. Importantly, this does not mean knowledge is invalid—but that it must be understood as relational and contingent. Fermentation embodies this idea in practice. It is a process where knowledge is deeply situated in the body, in place, in temperature, in timing. It cannot be fully written down or transmitted through standard recipes. Each fermentation is different. Each brew responds to conditions that shift daily: humidity, air, yeast behavior, the feel of the container. Brewers must rely not on fixed formulas but on experience, intuition, and responsiveness. In this way, fermentation offers a powerful metaphor and method, for feminist ways of knowing.

This kind of relational knowledge resonates strongly with the feminist value of care. Care, in feminist thinking, is not just an emotion or private duty. It is a political and epistemological practice: a way of paying attention, being responsible, and staying with complexity (Puig de la Bellacasa, 2017). In fermentation, care means more than hygiene. It means noticing when the yeast behaves differently, responding gently to signs of change, and being willing to discard or rescue a batch depending on how it “feels.” This work is time-consuming, uncertain, and rarely celebrated, but it is central to how good beer is made.

Importantly, feminist epistemology also allows us to hold tension rather than resolve it. The concept of competitive collaboration fits beautifully within this frame. Feminist theory reminds us that knowledge and relationships do not need to be harmonious to be meaningful. They can be conflicted, uneven, and still deeply connected. In the brewery, the yeast does not obey the brewer, and the brewer cannot fully surrender to the yeast. They push against each other, and through this push, something new is made. This idea connects to the feminist concept of intersectionality, first introduced by Kimberlé Crenshaw (1989). Intersectionality highlights how identities such as gender, race, class, and ability do not operate separately but interact to produce layered forms of privilege and oppression. Later scholars, such as Cho, Crenshaw, and McCall (2013), emphasized that intersectionality is not only a theory of identity but also a method of analysis. It asks us to examine how systems of power overlap and shape each other.

While feminist theory often highlights how multiple systems of power intersect to shape lived experience, in this thesis I focus more closely on how care and situated knowledge unfold in practice—particularly in the bodily and temporal dynamics of fermentation. This complexity is not a problem, it is the reality of life in a shared, interdependent world. And it is precisely this interdependence that fermentation makes visible. As Lauren Fournier (2020) argues in her project *Fermenting Feminism*, fermentation is not just a metaphor. It is a material, political, and ethical practice that teaches us about survival, transformation, and cohabitation. Her work brings together artists, writers, and scholars to explore how fermentation can inspire new ways of living with difference. It is slow. It is messy. It resists control. And for that reason, it offers a model for feminist ethics: grounded in attention, vulnerability, and change. In this view, fermentation is not a clean process. It is not linear. It bubbles, spills, fails, and recovers. It is fermentative: unstable, embodied, and relational. It teaches us that knowing is not about mastery but about response-

ability (Haraway, 2016), the ability to respond to others, human and nonhuman, even when we do not fully understand them.

Thus, when I describe the brewer-yeast relationship as competitive collaboration, I do not mean to resolve it into a simple binary of conflict versus harmony. Rather, I mean to describe a form of co-existence shaped by tension, where care and control, attention and resistance, presence and distance all operate at once. Feminist theory gives us the language to recognize this kind of relationship. Not as a failure of clarity, but as a space where knowing and relating happen together.

The Brewing Site as a Multispecies Environment

The brewery sits at the edge of a wetland park in central Mile, Yunnan. When I arrived in early summer, the air was thick with humidity, and the trees outside had already formed a dense canopy. Wind moved through the space in a slow, steady direction, from the back door, across the floor, and out through the front gate. Mile, located in the southeastern border region of China, spans a remarkably diverse landscape, from river valleys below 1,000 meters to mountain ranges above 2,000 meters in elevation. This ecological range allows both subtropical and high-altitude species to thrive. From the brewery, it takes only an hour by car to reach nearby mountain villages, where local farmers cultivate fruit trees and grains on terraced hillsides. The brewers told me one reason they chose this location was its proximity to high-quality raw ingredients. Highland apples and honey are among Mile's finest local products.

The first time I entered the brewery, I was struck by the smell—sharp, clinical, unmistakably chemical. Ethanol clung to the concrete floor and the metal fermentation tanks. I

had expected the scent of barley or hops, or at least something sweet. But unless it was the first day of ferment, those smells were rare. They disappeared into the disinfectant. The brewery had two floors, but almost everything related to brewing took place downstairs. It made sense: the equipment was too heavy to move up, and too wet to store above. Tanks, raw ingredients, hoses, and all of us, including the workers, the heat, and the microbes, shared the same level.

The ground floor was open from north to south, with connected windows on both sides that allowed light and air to pass through. Three 3,000-liter stainless steel fermentation tanks stood in a row on one side, wired to a control box that monitored temperature. On the other side, two shorter tanks of the same capacity surrounded a small platform that could hold two people. These were used for boiling and mashing. Behind the platform was a small room with a grain crusher, where grains were milled before entering the mash tun. Alongside this equipment, there was a small trolley carrying a pressure pump and two meters of tubing, plus a 20-liter bucket for disinfectant. This trolley was used to pump water and sanitize every corner of the space. I also noticed many trenches on the floor, draining toward the sewer. That was the entire downstairs. As the brewer explained this seemingly complex system to me, I was overwhelmed.

Figure 1

First Floor of Brewery

He reassured me that the process was actually simple, consisting of only four steps: crushing, mashing, boiling, and fermentation.

“First, we mix the malt and other grains with hot water. That pulls out the natural starches and gets them into the water. The enzymes in the grains then break those starches down into sugars. Those sugars are what the yeast will eat later to produce alcohol and carbon dioxide.” He went on, “After that, we add more water to help separate the wort, the sweet liquid, from the grain husks and other solids. Then we transfer the wort to a big tank and boil it. That stops the enzymes from doing anything more, locks in the sugar level, and

also sterilizes the liquid.” Finally, he said, “Once the boiling’s done, we cool the wort and move it to the fermentation tank. That’s when the yeast kicks in. With all that sugar from the malt, it gets to work.”

He continued,

“That’s what we learned in brewing school. But in reality, every step has too many variables. For example, in the crushing stage, some brewers prefer rolled oats over crushed ones. Oats are high in starch, and crushing them can overly gelatinize the content. Using rolled oats becomes a compromise: it preserves some starch for fermentation while avoiding excessive sugar release during the boil. These choices come from practice, each recipe calls for different ingredients, treatments, temperatures, and techniques.”

“It’s like a Chinese recipe,” he said, smiling.

“There’s no ‘one teaspoon’ or ‘30 milliliters’, just ‘some salt’ and ‘more sugar than salt.’”

This way of thinking made sense to him, and to me too.

The brewer I spoke with was not particularly fond of long conversations, so he invited me to begin brewing with him right away. And so, a typical brewing day began. My first experience was with a sour saison, also known as a farmhouse ale. Originally from France, this type of beer is marked by a sour flavor that comes from spontaneous fermentation. Spontaneous fermentation was their signature. No commercial yeast, no sealed fermenters. Just air, water, and time. During

the first days of fermentation, the wort was left open to the environment. The brewers stirred it by hand, every day, to invite whatever was in the air to join the process.

Figure 2



Outside of Brewery

I was told that wild yeast could not be controlled, but it could be “listened to.” The main brewer, he said,

“*Saccharomyces cerevisiae*, *Lactobacillus*, and acetic acid bacteria, each played a different role. *Saccharomyces* was the workhorse. It eats sugar, and gives us alcohol.”

One brewer explained that *Lactobacillus* played a quieter role in the process. It contributed a gentle sourness that helped prevent the growth of unwanted microbes, subtly protecting the beer from contamination. Acetic acid bacteria were less predictable, but just as important. They were responsible for producing volatile compounds that shaped the aroma, giving each batch its own distinct character. At times, their presence could be too sharp and overwhelming, but in the right balance, they added a deep and complex layer to the final flavor.

“You have to know when to stop.”

They told me. They believe these distinctions weren’t taught in school although they had system education about it. They keep telling me, they learned by repeated contact, by smelling, tasting, adjusting, remembering.

They described fermentation not as a formula, but as a process of attention.

“Here in Mile,” one said, “the air carries different things.”

He meant it literally: the wind here brought in pollen, tree fibers, even smells from the mountains.

“After the rain,” he added, “it’s the best. The yeast wakes up.”

It was about what was present in the air, what could feed, carry, or invite microbes to land. I began to understand that air was a selective carrier. This awareness shaped every decision. In the early days of a new batch, we stirred more often, letting the wort meet the air. Sometimes we opened the windows a bit wider. Sometimes we waited for the late afternoon breeze. None of

this was written down. But the brewers moved with a practiced intuition, not trying to control the process, but to accompany it.

At the end of that day, he told me something I would remember:

“In brewing, the most important thing is cleaning. A good brewing day should be 70% cleaning.”

I asked why. He said it was about creating a baseline, ensuring the beer was safe to drink. I asked, “What about the raw materials?” He agreed they were important, but pointed out that beer, as a highly commercialized industry, relied on stable supply chains. Barley, wheat, oats, rye, hops, yeast, each had regular suppliers. Most brewers stuck to fixed vendors and varieties to ensure product consistency, adding only a small number of local ingredients to express regional character. That was another reason he liked Mile:

“There’s so much local terroir to work with here.”

Still, this openness had its boundaries. Cleaning took up most of our time. Before each fermentation, every vessel was scrubbed. Tanks were wiped with 75% alcohol. Floors were rinsed and dried. We were preparing the space to receive the cooled sour Saison wort, making sure every surface the wort might touch was clean. During a short break, I asked if such thorough cleaning killed all bacteria, even the helpful ones. A brewer replied without hesitation:

“It’s not about killing everything. It’s about preparing the space for the right ones.”

He believed that some beneficial microbes might be lost in the process—but if they were not present at the right time, then they were not the right ones. Only those microbes that

appeared at the right time and in the right place belonged in the tank. Cleanliness, it was an invitation with boundaries. Let the wild in, but not the wrong kind.

Figure 3



Cleaning Sewer

I understood this boundary again one afternoon. The temperature inside the brewery rose quickly. Foam in one tank swelled too fast and began to overflow. No alarms sounded. No instruments were touched. The brewer walked over, looked into the tank, and opened both windows wide. I stood nearby, stirring. A sharp, sour smell rose from the surface.

“Let it run for a bit,” he said calmly.

Nothing dramatic happened, and that was the point. Airflow was the response. Waiting was the method. I realized fermentation was a sequence of gestures. Part habit, part intuition, part microbial negotiation.

I was once encouraged to wash my hands with fermented sediment, even to try it on my face. Each day after fermentation began, we had to discharge sediment from the tank. Although raw materials were filtered before entering the tank, fermentation constantly produced active and inactive yeast. This yeast looked like curdled milk—soft clumps mixed with fine sediment. I was responsible for removing this material. The yeast felt different every day. Sometimes smooth and fragrant, sometimes sharp and tingly. The brewers noticed these differences too. They touched the surface, smelled the foam, checked the color of the residue. Each morning, they siphoned off the settled yeast and read its density, scent, and hue. It wasn’t language in the usual sense. But it was a way of reading.

Figure 4



Yeast

Fermentation and cleanliness, two forces that are both similar and in conflict. The yeast was not an ingredient. It was a presence, unruly, invisible but vital. The air was not neutral. It was active. Cleanliness was not absence. It was a way of shaping who could enter and stay. And the brewers, including myself as a temporary one, were not only making beer. We were managing a fragile and collaborative space, where humans and microbes worked side by side, sometimes in agreement, sometimes not, but always in relation.

What I observed in this brewery was not a rejection of microbial control, nor a full embrace of wildness. It was something in between, a way of working with microbes that neither assumed full knowledge nor ceded full agency. Some scholars might call it “post-Pasteurian”: not sterilized, not ungoverned, but situated, practiced, and constantly negotiated. The yeast was not a tool, and the brewer was not a master. Their relationship was shaped by air, timing, and sensory judgment, by waiting and reacting, rather than commanding. In this sense, the brewery was not only a site of production. It was a space where human and nonhuman actors entered into ongoing conversation, mediated by care, attention, and sometimes, disagreement.

Robbing the Definition Power of Cleanliness

It began with a drive into the mountains, toward a village nearly 3,000 meters above sea level. The road wound sharply between terraced fields, cutting through mist that hovered just above the treetops. The brewer sat beside me in silence, holding a plastic fermentation bucket between his knees. It had been cleaned and sterilized the night before, first with boiling water, then alcohol, then boiling water again. Inside was a few liters of cooled wort. He said nothing during the drive, as if speaking too soon might interfere with what was about to take place.

When we arrived, he placed the bucket in a small courtyard, just beside a row of weathered beehives stacked unevenly under a wooden eave. The hive boxes were scratched and worn, with lids slightly ajar. The courtyard belonged to an elderly woman who sold wild honey gathered from the surrounding hills. She didn't keep bees, she received them. The honey came to her through wind, flowers, and accidents.

Figure 5



Beehive and bees

“The yeast here,” the brewer explained, “should be strong. It’s used to surviving.”

I had a brief conversation with the beekeeper lady. She told me that if we had come a year earlier, there would have been many more bees and hives. During the 2025 Chinese New

Year, a group of children set off firecrackers on the dry winter grass nearby. A fire broke out, destroying many hives. Some bees were burned, and many did not return during summer. She looked a little sad as she spoke. Some of the hives, she said, had belonged to her father. She felt she had failed to protect them, and the bees.

Figure 6



Beekeeper lady

The wort sat in that courtyard for three days, exposed to the open air. It was covered only with a cloth to keep out leaves and large insects. No seal, no tape, no thermometer. Just air and time. Wind passed steadily from the west, carrying the scent of pine, soil, and sun-warmed stone. The land here was wide open, halfway up the mountain, surrounded by forest and a few scattered

homes and fields. The brewer didn't test anything. He simply returned every evening to look. I followed him each time, and noticed the liquid dropping slowly in volume. Small bubbles began to form around the edges of the bucket. Every day, new insects drowned in the liquid, and the brewer gently removed the bees' bodies with a cloth.

When we came back to collect the sample, the wort had turned faintly cloudy. A thin, white film had formed across the top, barely visible unless the light struck it at the right angle. Its smell had changed. It was clean, but carried a sharp, electric sourness. The brewer dipped his finger into the liquid, touched it to his lips, and didn't speak. Then he nodded once, and we lifted the bucket into the car. The descent back to the brewery took less than an hour, but the shift in air felt heavier with every kilometer. The village had been high, dry, sparse. The brewery sat at 800 meters, low, humid, enclosed. It felt like returning to sea level after standing on a ridge. The yeast had come from bees, from altitude, from silence. Now it would have to adapt to steel, to warmth, to fluorescent light.

Back at the brewery, the brewer placed the bucket into a temperature-controlled chamber. It was a standard protocol he used, a softly lit incubator set to moderate, steady heat. He checked the lid, the seals, the switches.

Then the cleaning started again.

The table, the tank, the tools, the floor—everything was steamed twice. Even the thermometers, one outside the tank and one submerged inside, were rinsed with alcohol and dried with bleached cloth. The smell of ethanol lingered for hours. The walls shone under a new layer of invisible hygiene.

Figure 7



Steaming Every Surface

We stayed at the brewery for three days. The wort rested quietly in a 100-liter stainless tank, sealed with a loose-fitting lid. Every few hours, the brewer returned to steam the table, replace the alcohol cloths on the floor, and rinse the thermometers, even if no one had used them. He called this phase “inoculation”, a stage where the wild yeast, introduced in the mountains, would be mixed with a larger volume of wort to activate fermentation.

“This is the most unpredictable part,” he said. “If it works, we’ll get a hundred liters of great beer. If not, we’ll lose all the time, water, ingredients, and disinfectant.”

He wore latex gloves to check the temperature. One glance at the digital display, then back to wiping the tank’s edge in quiet, circular motions. He spoke softly, as if sound itself could interfere. The surface of the wort remained still. No foam, no haze, no rising gas. The only change was a faint sourness replacing the original grain smell.

“I’ve cleaned everything I can,” he said one afternoon. “Now we wait.”

He did not increase the temperature beyond a preset range. He added no nutrients. He avoided opening the lid.

“I don’t want to interrupt its own logic,” he explained. “This yeast isn’t from here. It might not like what we usually do.”

He suspected that due to the altitude shift, the yeast might need more time or higher temperatures to activate. But increasing the heat too quickly could kill it. So, he chose to wait. The yeast had been collected at 3,000 meters and was now in a room below 800. The drop in

elevation brought changes in temperature, air pressure, and humidity, none of them visible, but all of them real.

Figure 8



Sample Observation

The yeast had joined the wort with no plan. Now, it was quiet.

By the second day, the liquid became slightly cloudy. A thin, uneven film appeared, with a texture like diluted milk. On the third morning, small bubbles collected along the rim from one corner, where a faint thread of foam had formed.

The smell shifted again. It no longer carried a general sourness, but something sharper, like overripe fruit left on warm concrete. The brewer leaned in, inhaled lightly, and nodded. It's waking up, he said. After that, he stopped touching the tank. No more cleaning. No more rinsing unused tools. The gauze on the table stayed unfolded. The ethanol bottles remained closed. His posture changed, too. He no longer stood alert. He sat by the wall, watching quietly. He told me the inoculation was complete, and that he had done everything he could. I was confused—it didn't yet look like beer. He explained:

“All the cleaning was to give the collected yeast a good growing environment, to help it fight off dangerous bacteria. Environmental yeast, by definition, depends on its environment. But as you can see, I can't predict how it will grow. It follows its own pattern. So the only thing I can do is make sure that no contaminating microbes get in through by cleaning.”

And after that, he really did stop. Yet the fermentation didn't immediately progress either. The sourness, the color, both stopped changing. Only the bubbles continued, slowly rising to the surface. These small signs seemed to represent something. I saw them as a kind of polite boundary.

Cleanliness, once proactive, became observational. The yeast had not been instructed. It had responded to conditions we couldn't fully perceive, perhaps temperature had stabilized in a way it liked, or the humidity crossed some unseen threshold. There was no way to know. But something had shifted. And that shift redefined what “clean” meant in that room. For the brewer, cleanliness meant wiping, sealing, isolating risk, a way of holding the space steady. For the yeast, cleanliness meant time without interference, air without additives, warmth without

pressure. Their definitions did not match. But they did not clash either. They met in a kind of standoff. The brewer cleaned. The yeast delayed. The process began, not through command, but through parallel endurance.

Figure 9 and Figure 10



Bubbling in Fermentation Tank

In the following days, the brewer came only once a day—just to check. He told me that wild yeast has its own path of development. All we can do is taste, use flavor as a way to sense the yeast's direction. In textbooks, a typical sour Saison should have balanced sourness, fruitiness, and notes of citrus and pepper. Based on his experience, he predicted this batch would be lighter and less acidic. Because fermentation took longer to start, the alcohol content would be

lower, and the flavor more gentle. He was right. This batch had a soft citrus aroma and a rounded, mild body.

In the end, the fermentation moved forward. Bubbles became regular, the liquid darkened, and the scent softened. The yeast had settled in, or perhaps, the space had settled around it. The brewer didn't speak of success or failure. He just recorded the date on the side of the tank and moved on to the next task.

But I kept thinking: so much had been done, and not done, for this to happen. Control hadn't been established. It had been suspended. Cleanliness hadn't been enforced. It had been negotiated, moment by moment, through cloths and lids, through temperatures and delays, through silence and smell. No final agreement was reached. But something like alignment emerged—tense, partial, incomplete.

What unfolded here was not harmonious collaboration. It was a competitive standoff, uneven and unresolved. And yet, fermentation happened. Not because one side won, but because both stayed.

Findings: Competitive Collaboration as a Mode of Human-Microbial Interaction

The more time I spent inside the fermentation's space, the more I realized that what I was witnessing could not be described as either harmonious cooperation or direct control. The relationship between the brewer and the yeast was structured around uncertainty, managed but never eliminated. They needed one another, but did not move in synchrony. They remained in contact, but not in agreement. In my case, fermentation delayed for three days after the inoculation of wild yeast collected from a mountain village. The brewer responded not by

interfering with the yeast, but by intensifying the cleaning of the surrounding space: wiping the tank walls, adjusting the temperature, sterilizing tools that had not been touched. Meanwhile, the yeast showed no visible signs of activation. It was not refusing, but it was not cooperating either. When fermentation eventually began, it did so on its own terms, and the brewer quietly adjusted his rhythm in response. Moments like these repeated throughout my fieldwork. Across different fermentation trials, I observed that collaboration between brewers and microbes did not emerge from shared intention or transparent communication. It was forged through asymmetrical persistence: both sides remained active in the same space, testing, resisting, adapting, but never fully conceding.

This observation led me to develop the concept of competitive collaboration. As the central I use this term to describe a relational mode in which both human and microbial actors pursue their own priorities, cleanliness, stability, activation, survival, while remaining entangled in a shared process. By highlighting the persistent tension rather than harmony, competitive collaboration provides an innovative lens that enriches existing multispecies ethnographic frameworks, bringing everyday microbial negotiations into clearer theoretical focus. It echoes what Paxson (2013) calls a “post-Pasteurian” approach: one that embraces microbial agency not through eradication, but through responsiveness and negotiation. And as Anna Tsing (2015) reminds us, friction itself can be generative, it produces movement, not despite difference, but because of it.

But while Paxson and others describe post-Pasteurian food practices as open to microbial participation, they often still presume that a form of managed harmony will eventually emerge. Competitive collaboration departs from this expectation. It captures moments where human

actors, like brewers, do not simply “allow” microbial agency but are compelled to contend with its unpredictability. In these situations, yeast does not merely add complexity, it introduces tension. It resists timing, alters outcomes, and imposes its own conditions of cooperation. What makes this collaboration “competitive” is not just that humans and microbes have divergent goals, but that they attempt to shape the same environment in incompatible ways. The brewer sterilizes the fermentation vessel to prepare for activation; the yeast delays that activation, testing whether the vessel is suitable. Both act within the same material context, but according to different logics. This overlap without alignment is the defining feature of competitive collaboration. Importantly, this is not a temporary failure of coordination. It is the relationship. The brewer continues to clean; the yeast continues to adapt. Neither yield, yet both remain. Collaboration, here, is sustained not by consensus but by persistence within difference, a relation maintained precisely because neither side gives up its terms.

In this concept, fermentation emerge as a place that felicitate the relationship happened. This form of interaction takes place not in abstraction, but within a specific material and temporal context. As suggested earlier in this thesis, fermentation is not only a biochemical reaction or a production technique, it is also a space. A space that holds together difference, allows tension to persist, and invites relation without guaranteeing consensus. In this sense, fermentation offers more than flavor or transformation. It becomes an environment in which human and microbial agencies can confront, negotiate, and coexist.

What makes fermentation unique is that it does not resolve conflict. It contains it. Through its timing, uncertainty, and sensory demands, fermentation slows the process of domination. It does not allow one agent to easily overpower the other. The brewer may clean the

tank, but the yeast still decides when to activate. The yeast may respond to the temperature, but it cannot control how long the environment will remain stable. Their encounter unfolds in a shared yet unstable atmosphere, where neither side can fully predict the other.

Knowledge in the fermentation was not written, spoken, or tested. It was touched, sniffed, observed. In these exchanges, it was not only the brewer observing the yeast. The yeast, in turn, responded to the conditions prepared by human labor: to temperature, airflow, surface cleanliness, light. It did not speak, but it adjusted. The rate of bubbling changed. The liquid surface shifted. Fermentation accelerated, or stalled. This back-and-forth formed a kind of feedback loop: not technical, but bodily. It was a loop made of foam, acidity, waiting, wiping, and remembering. What the brewer learned was not just taught. It was sensed over time, through failures and revisions. And what the yeast offered was not data. It was behavior. These practices resonate with feminist theories of knowledge as situated and embodied. Rather than separating the knower from the known, they reflect a process in which knowledge is shaped by involvement, attention, and the willingness to be affected. The brewer does not observe from a distance; he participates through care, maintenance, and repetition. As feminist epistemologists such as Haraway and Puig de la Bellacasa have shown, knowing is not neutral, it is always embedded in relations that are uneven, partial, and alive.

Such interactions do not fit easily into conventional ideas of knowledge production. There were no variables to isolate. No outcomes to guarantee. Instead, there was attunement: an ongoing calibration between species, carried out through smell and touch, delay and repetition. In this shared space, knowledge was not extracted. It was composed, moment by moment, across porous boundaries. What unfolded in the fermentation was not a partnership in the usual sense.

The brewer did not yield control, and the yeast did not respond obediently. Their actions overlapped without aligning. The result was neither failure nor harmony, but something more unstable: something that held its shape precisely because of the tension inside it.

To collaborate without consensus is not failure. It is a different way of continuing. In fermentation, what holds human and microbial actors together is not shared understanding, but sustained responsiveness. Their actions remain uneven, sometimes even resistant to one another, yet neither steps out of the relation. This staying, in repetition, in adjustment, in attention, is what gives form to what I call competitive collaboration. It is not a model of agreement. It is a method of enduring difference. In the context of fermentation, it becomes possible to think of knowledge not as clarity, but as commitment. A practice that allows contradiction to persist, and makes space for life to unfold without resolution.

Discussion

Throughout this research, fermentation emerges as a distinctive epistemological space where multispecies relations unfold through continuous yet subtle negotiations. Rather than existing as stable, predefined interactions, relationships between human brewers and microbes materialize through practices marked by friction, uncertainty, and responsive adjustments. In foregrounding the concept of competitive collaboration, this thesis moves beyond binary categorizations of human control versus microbial autonomy, proposing instead that interaction is inherently characterized by tension and mutual influence.

Competitive collaboration enriches multispecies ethnography by emphasizing that coexistence need not be harmonious or clearly reciprocal to be meaningful. This research thus

extends Paxson's (2013) notion of the Post-Pasteurian landscape by explicitly highlighting how microbial agency resists simple inclusion or exclusion, instead continuously redefining the boundaries and meanings of cleanliness and control. This challenges anthropological narratives that either romanticize nonhuman actors or implicitly maintain human mastery. My research demonstrates how multispecies interactions are characterized by dynamic tensions, moments of resistance, delay, and negotiation, that are intrinsic, productive, and necessary for coexistence rather than indicative of its breakdown.

Additionally, this analysis contributes significantly to feminist epistemologies of situated and embodied knowledge (Haraway, 1988; Puig de la Bellacasa, 2017). In the brewery, knowledge production is neither unilateral nor purely cognitive but deeply embodied and materially enacted. Brewers learn not simply through instruction or technical mastery, but through a practice of attunement, responsiveness, and sensory interaction with microbial life. Knowledge here is emergent, partial, and deeply contextual, requiring continuous recalibration rather than straightforward acquisition. In this way, competitive collaboration not only describes interaction but offers a model for understanding knowledge as a profoundly relational and contingent practice.

The implications of recognizing competitive collaboration as a mode of multispecies interaction are broad. By decentering human intentionality and acknowledging microbial participants as co-creators of knowledge, this concept prompts rethinking beyond brewing contexts. In agriculture, for instance, conventional approaches typically frame farmers' relationships with pests, weeds, and beneficial microbes in terms of domination and eradication. Competitive collaboration challenges this perspective, suggesting instead that farmers are

constantly engaged in subtle negotiations with nonhuman actors. This reframing invites a critical reassessment of agricultural practices, moving away from unilateral control towards methods that recognize and accommodate the persistent presence and agency of nonhuman species. Such a shift emphasizes responsiveness, adaptability, and coexistence, aligning agricultural practices more closely with sustainable and regenerative principles.

Similarly, in medical anthropology contexts, human bodies continuously interact with complex microbial communities. Traditional medical paradigms often conceptualize microbial life in binary terms: beneficial or harmful, present or absent. Competitive collaboration complicates this simplified view by emphasizing that health and illness are emergent states produced through delicate negotiations within multispecies communities. From this standpoint, medical practices and public health strategies might prioritize care, vigilance, and attunement rather than sterile control. This would entail a profound transformation in healthcare philosophy, promoting approaches that actively engage with microbial presence as dynamic and relational, rather than strictly pathogenic or beneficial.

Competitive collaboration also opens pathways for comparative ethnographic exploration, providing a robust analytical framework to understand how multispecies tensions manifest differently across diverse cultural and ecological settings. Comparative inquiries could examine how urban spaces shape human-microbial negotiations compared to rural environments. Do cities, with their distinct microbial ecologies and sanitation practices, foster different modes of competitive collaboration than agricultural or wilderness areas? Moreover, how do these interactions vary across global fermentation traditions, each with its own historical, cultural, and

ecological contexts? Addressing these questions could reveal both the cultural specificity and potential universality of multispecies tensions and negotiations.

Conclusion

This thesis explored the intricate and dynamic interactions between brewers and microbial life within a natural brewery in Yunnan, China. By employing multispecies ethnography and feminist epistemology, I introduced the concept of competitive collaboration as a distinctive mode of human-microbial relationship, marked by persistent negotiation, tension, and responsiveness. Rather than achieving full harmony or succumbing to conflict, humans and microbes coexist in continuous dialogue, each influencing and responding to the other's presence, decisions, and actions.

Fermentation, as both a material process and epistemological metaphor, has shown us that knowledge creation is not merely an act of mastery or extraction but one deeply rooted in embodied experience, situated practice, and sensory attentiveness. Within this fermentative space, cleanliness emerged as a contested yet pivotal category, dynamic, relational, and continuously renegotiated. Cleanliness was not merely a sanitized standard but a flexible boundary, managed through careful acts of inclusion and exclusion. This relational approach to cleanliness reflects broader tensions in human-nonhuman interactions, particularly around issues of control, care, and coexistence.

Through my immersive fieldwork and reflections on positionality, I highlighted how multispecies interactions, such as those between brewers and yeast, mirror broader social relations and power dynamics. Recognizing my dual position as both insider and outsider

allowed me to observe these subtle interactions without simplifying them into binary narratives. Competitive collaboration provides a nuanced lens, challenging anthropocentric models and urging anthropologists to remain open to ambiguity, partiality, and the persistence of difference.

By bridging feminist epistemology with multispecies ethnography, this thesis contributes to ongoing anthropological discussions about nonhuman agency, interspecies communication, and the politics of knowledge production. Ultimately, competitive collaboration calls for an anthropology that remains attentive to complexity and committed to relational practices, an anthropology open to continuous transformation, shaped by attentiveness, humility, and the capacity to engage meaningfully across species boundaries.

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